

# SPAM FARM

\* Indicates required question

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1. Email \*

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2. Roll number \*

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3. Password \*

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## Question

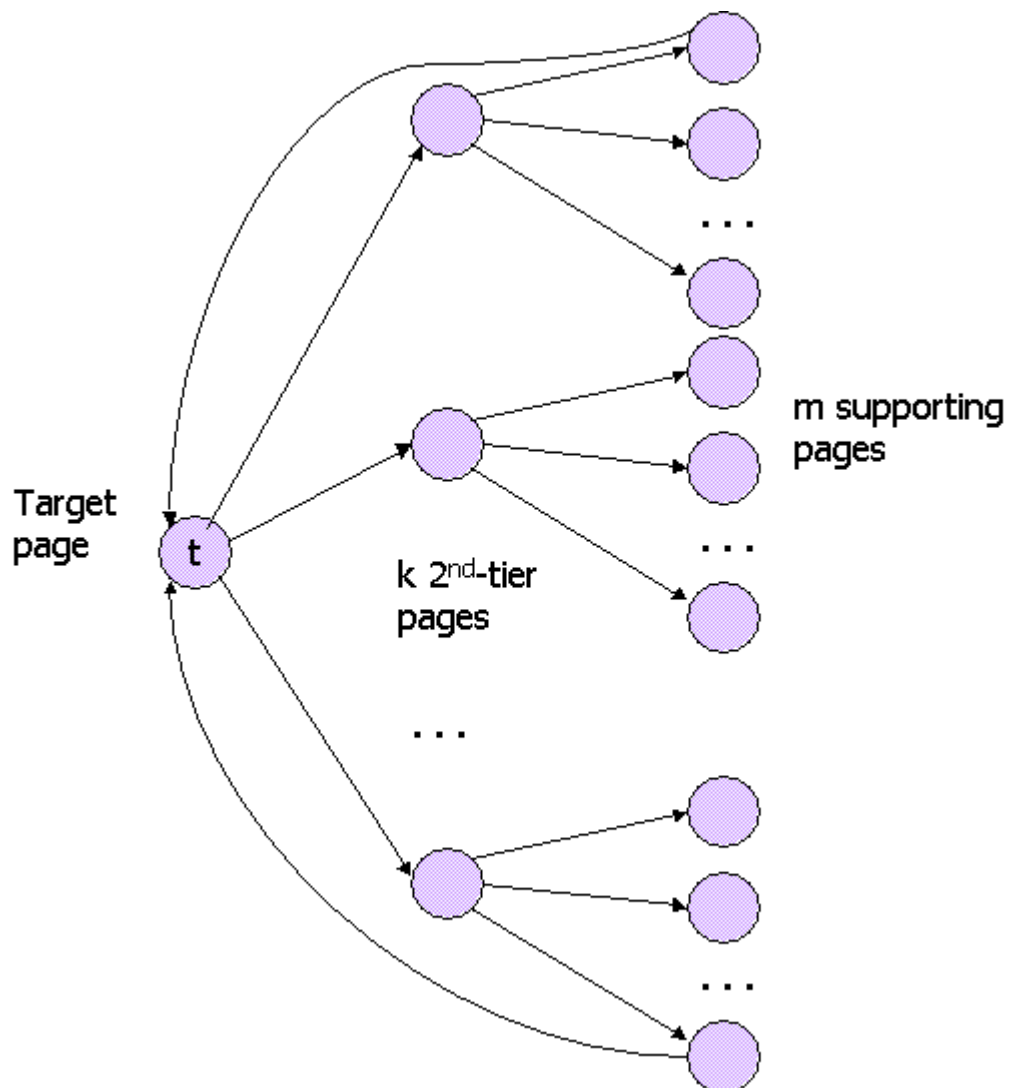
The spam-farm architecture as discussed in the class suffers from the problem that the target page has many links --- one to each supporting page. To avoid that problem, the spammer could use the architecture shown in the figure.

There,  $k$  "second-tier" nodes act as intermediaries. The target page  $t$  has only to link to the  $k$  second-tier pages, and each of those pages links to  $m/k$  of the  $m$  supporting pages. Each of the supporting pages links only to  $t$  (although most of these links are not shown). Suppose the taxation parameter is  $\beta = 0.85$ , and  $x$  is the amount of PageRank supplied from outside to the target page. Let  $n$  be the total number of pages in the Web. Finally, let  $y$  be the PageRank of target page  $t$ . If we compute the formula for  $y$  in terms of  $k$ ,  $m$ , and  $n$ , we get a formula with the form

$$y = ax + bm/n + ck/n$$

Note: To arrive at this form, it is necessary at the last step to drop a low-order term that is a fraction of  $1/n$ . Determine coefficients  $a$ ,  $b$ , and  $c$ , remembering that  $\beta$  is fixed at 0.85. Then, identify the value, correct to two decimal places, for one of these coefficients.

## Spam Farm



## 4. Options

1 point

Check all that apply.

- ☐  $b = 0.33$
- ☐  $c = 0.28$
- ☐  $a = 3.07$
- ☐  $b = 0.28$
- ☐  $a = 2.59$

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